

Task 72: Reverse Bits

PROBLEM DESCRIPTION

Reverse bits of a given 32 bits unsigned integer. Note: In some languages, such as Java, there is no unsigned integer type. In this case, both input and output will be given as a signed integer type.

JAVA SOLUTION

```
public class Solution {
    // you need treat n as an unsigned value
    public int reverseBits(int n) {
        int result = 0;
        for (int i = 0; i < 32; i++) {
            // Shift result to the left by 1 to make space for the new bit
            result <<= 1;

            // If the rightmost bit of n is 1, add 1 to the result
            if ((n & 1) == 1) {
                result++;
            }

            // Shift n to the right by 1 to process the next bit
            n >>= 1;
        }
        return result;
    }
}
```